

FocusFlow — Complete Requirements & Feature Specification

1. Project Definition

Project Name

FocusFlow – Interactive 7-Day Focus Development Web Application

Project Type

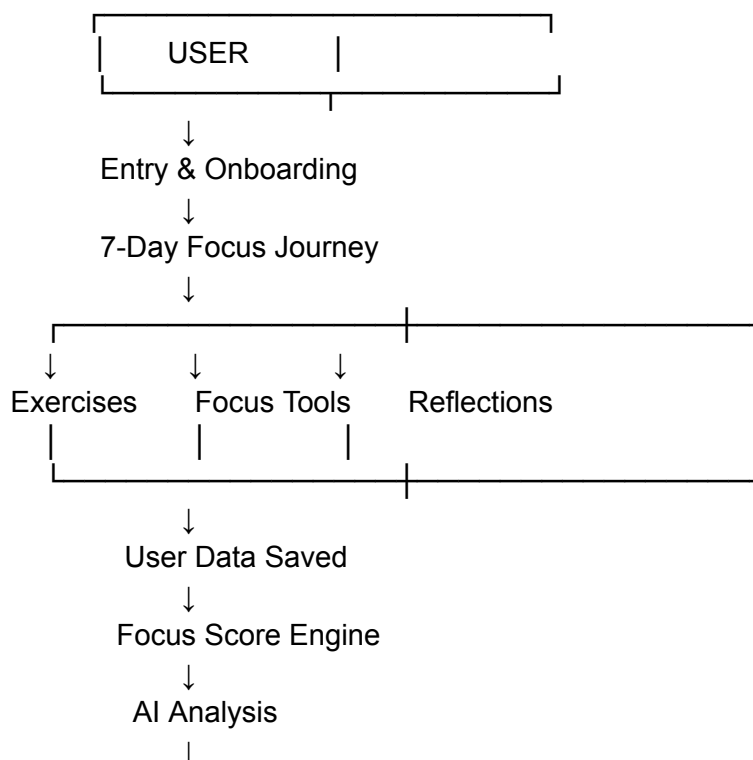
University Web Application / Software Engineering Project

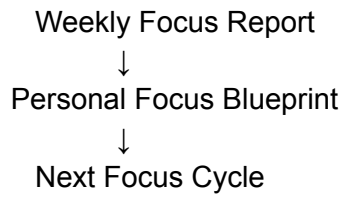
Core Idea

FocusFlow converts the fixed 7-day methodology of *Powerful Focus* into an interactive web application.

The system guides users through seven structured days, collects their responses and activities, measures their focus-related behavior, and generates a personalized report using an LLM API.

Main System Flow





2. Main System Modules

Your workflow diagram can be divided into **five major system areas**, exactly as shown in your diagram.

Module	Purpose
1. Entry & Onboarding	Introduce and prepare the user
2. Core 7-Day Journey	Deliver the fixed seven-day program
3. Interactive Modules	Provide practical tools
4. Scoring & Feedback	Calculate Focus Score and reports
5. Backend Services	Handle data, AI, analytics and persistence

3. Module 1 — Entry & Onboarding

This is the first part of the user's experience.

3.1 Landing Page

The landing page should explain:

- What FocusFlow is
- Why focus matters
- How the 7-day journey works
- What the user will achieve
- How AI is used
- Start Journey button
- Login button
- Sign-up button

Example

FOCUSFLOW

Build Better Focus
in 7 Days

Understand what matters.
Remove distractions.
Focus on high-impact work.

[Start 7-Day Journey]

3.2 User Registration

The user can create an account.

Required information

- Name
- Email
- Password

Functional Requirements

FR-01: The system shall allow users to create an account.

FR-02: The system shall validate registration information.

FR-03: The system shall prevent duplicate email registration.

FR-04: The system shall securely store user credentials.

4. Login

Existing users can log into FocusFlow.

Functional Requirements

FR-05: The system shall allow registered users to log in.

FR-06: The system shall verify user credentials.

FR-07: The system shall maintain the user's session after successful login.

FR-08: The system shall provide logout functionality.

5. Onboarding

After registration, the user receives a short introduction.

The system explains:

Welcome to FocusFlow.

You will complete a 7-day journey.

Each day contains:

- ✓ A short lesson
- ✓ An exercise
- ✓ Reflection
- ✓ Progress tracking

The system can also ask the user for:

- Current focus level
- Main reason for joining
- Current biggest distraction
- Primary personal goal

These become the user's initial baseline.

6. Module 2 — Core 7-Day Journey

This is the **heart of FocusFlow**.

The content is fixed.

The user cannot skip directly from Day 1 to Day 7.

Basic flow

Day 1



Complete



Day 2 unlocked



Day 3 unlocked

↓
...
↓
Day 7
↓
Final Analysis

Your diagram already shows this logic through:

Day 7 complete? → Yes → Generate Weekly Report

7. Day 1 — Discover Yourself

Purpose

Help users understand:

- Desires
- Strengths
- Abilities
- Passions
- Values

Feature

Interactive Self-Discovery Questionnaire

Example:

What do you really want from your life?

User's answer...	
------------------	--

Functional Requirements

FR-09: The system shall display predefined Day 1 questions.

FR-10: The system shall allow users to submit written answers.

FR-11: The system shall automatically save answers.

FR-12: The system shall allow users to edit answers before completing the day.

FR-13: The system shall mark Day 1 as completed after all required activities are submitted.

Data Produced

Self-Discovery Data



Desires

Strengths

Passions

Values

Abilities

8. Day 2 — Define Your Goals

Purpose

Help users identify meaningful goals.

Goal categories

- Career
- Financial
- Health
- Relationships
- Personal Development

Feature

Goal Clarity Dashboard

Example:

MY GOALS

Career

→ Become a Software Engineer

Financial

→ Save X amount

Health

→ Exercise 4 days/week

Personal

→ Read 2 books/month

Functional Requirements

FR-14: The system shall allow users to create goals.

FR-15: The system shall allow users to categorize goals.

FR-16: The system shall allow users to edit or remove goals.

FR-17: The system shall allow users to identify their priority goals.

FR-18: The system shall save goal information in MongoDB.

9. Day 3 — Goal → Action Planner

This converts goals into practical actions.

Example

GOAL

Become a Web Developer



Actions



Learn JavaScript



Build Projects



Learn Node.js



Build Full-Stack Project

Feature

Goal → Action Planner

Functional Requirements

FR-19: The system shall allow users to select a goal.

FR-20: The system shall allow users to create actions for the goal.

FR-21: The system shall allow users to mark actions as completed.

FR-22: The system shall display the relationship between goals and actions.

10. Day 4 — 80/20 Task Analyzer

This is one of your most interesting interactive features.

The user enters their tasks.

Example:

Task	Time	Impact
Learn JavaScript	2h	High
Build project	2h	High
YouTube	2h	Low
Facebook	1h	Low

The system analyzes the tasks.

Feature

80/20 Task Analyzer

Output

HIGH-IMPACT ACTIVITIES

- ✓ Building projects
- ✓ Learning JavaScript

LOW-IMPACT ACTIVITIES

- △ Random YouTube
- △ Social media

Functional Requirements

FR-23: The system shall allow users to enter tasks.

FR-24: The system shall allow users to specify estimated time.

FR-25: The system shall allow users to identify task importance/impact.

FR-26: The system shall categorize tasks according to their impact.

FR-27: The system shall identify high-impact tasks.

11. Day 5 — Distraction Management

Feature

Distraction Blocker / Focus Session

The user selects a task:

CURRENT TASK

Build JavaScript Project

50:00

[Start Focus]

During the session the application records:

- Start time
- End time
- Duration
- Task
- Interruptions

Functional Requirements

FR-28: The system shall allow users to start a focus session.

FR-29: The system shall provide a configurable focus timer.

FR-30: The system shall record completed focus sessions.

FR-31: The system shall allow users to record interruptions.

FR-32: The system shall track distraction events during focus sessions.

12. Say No Simulator

This is another interesting module from your workflow.

The system gives the user situations.

Example

Your friend asks you to attend a social event while you have an important deadline.

The user chooses or writes a response.

The AI can later analyze the response, but this can remain a secondary feature.

Functional Requirement

FR-33: The system shall provide predefined "Say No" scenarios.

FR-34: The system shall allow users to provide responses to scenarios.

FR-35: The system may use AI to provide feedback on the response.

13. Day 6 — Information Diet

The user records information consumption.

Example:

Activity	Duration	Useful?
YouTube	2h	No
Facebook	1h	No
Programming tutorial	1h	Yes
Research	30m	Yes

Feature

Information Diet Tracker

Functional Requirements

FR-36: The system shall allow users to record information activities.

FR-37: The system shall record time spent.

FR-38: The system shall allow users to identify whether information was useful.

FR-39: The system shall calculate useful vs unnecessary information consumption.

14. Day 7 — Review

Day 7 brings everything together.

The system displays:

YOUR 7-DAY JOURNEY

- ✓ Self Discovery
- ✓ Goal Clarity
- ✓ Action Planning
- ✓ 80/20 Analysis
- ✓ Distraction Management
- ✓ Information Diet

↓

REVIEW

↓

FINAL REFLECTION

The user provides a final reflection.

15. Module 3 — Focus Score Engine

Now we have the **0–100 Focus Score**.

The important design decision is:

The Focus Score is calculated by the application, not invented by the AI.

This makes it measurable and explainable.

Suggested Focus Score Components

We can use five components:

1. Goal Alignment

20 points

How clearly the user has defined meaningful goals.

2. High-Impact Work

25 points

How much of the user's recorded activity is focused on high-impact tasks.

3. Focus Consistency

20 points

Based on completed focus sessions and consistency.

4. Distraction Management

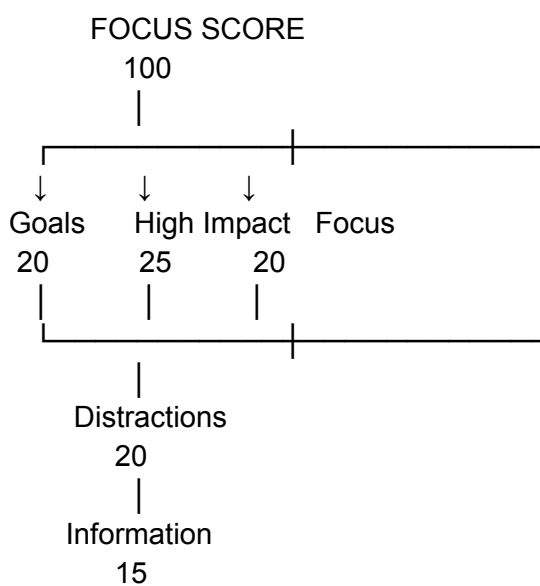
20 points

Based on interruptions and distraction activity.

5. Information Management

15 points

Based on useful vs unnecessary information consumption.



Total:

$$20 + 25 + 20 + 20 + 15 = 100$$

We can define the exact formulas later when we implement the scoring engine.

16. Focus Score Dashboard

The user sees:

YOUR FOCUS SCORE

72
/100

Goal Alignment	16/20
High-Impact Work	18/25
Focus Consistency	15/20
Distraction	14/20
Information	9/15

This gives the user a clear explanation instead of simply saying:

"Your score is 72."

17. Module 4 — AI Analysis

This is where your **LLM API** comes in.

The AI does **not** control the application.

Instead:

User Data



MongoDB



Backend collects relevant data



AI Prompt



LLM API



AI Analysis



Structured Report

↓
MongoDB
↓
User Dashboard

18. What Data Does AI Analyze?

The AI can analyze:

Day 1

- Self-discovery answers
- Values
- Passions
- Strengths

Day 2

- Goals
- Priorities

Day 3

- Goals → Actions

Day 4

- High-impact tasks
- Low-impact tasks

Day 5

- Distractions
- Focus sessions
- Interruptions

Day 6

- Information consumption

Day 7

- Final reflection

And the calculated Focus Score.

19. AI-Generated Weekly Report

The report could contain:

1. Executive Summary

A short overview of the user's week.

2. Goal Analysis

What the user appears to prioritize.

3. Strengths

Patterns identified from their answers.

4. Focus Patterns

What activities appear to receive the most attention.

5. Distraction Patterns

Potential recurring distractions based on recorded data.

6. High-Impact Activities

Activities that the user identified as high impact.

7. Information Consumption Analysis

Useful vs unnecessary information.

8. Recommendations

Practical suggestions based on the user's data.

9. Next Action

One concrete action the user can take.

20. Personal Focus Blueprint

This should be the **final output of the entire 7-day journey**.

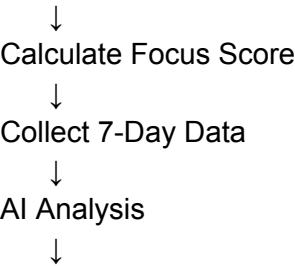
MY PERSONAL FOCUS BLUEPRINT	
MY MAIN GOAL	
Become a Software Engineer	
WHY IT MATTERS	
Personal growth + career	
MY STRENGTHS	
Learning + problem solving	
HIGH-IMPACT ACTIVITIES	
• Building projects	
• Programming practice	
DISTRACTIONS TO REDUCE	
• Random YouTube	
• Social media	
FOCUS SCORE	
72 / 100	
NEXT ACTION	
Build one project this week	

This is the feature I would emphasize during your presentation.

21. Module 5 — Weekly Report

After Day 7:

Day 7 Complete



Generate Weekly Report



Generate Focus Blueprint

The report should be stored so the user can view it later.

22. Backend Services

Your workflow diagram contains four major backend services.

API / App Server

Node.js + Express.js handles:

- Authentication
 - User requests
 - Daily progress
 - Goals
 - Tasks
 - Focus sessions
 - Reports
 - AI requests
-

User & Progress Database

MongoDB stores:

Users
Goals
Daily Answers
Tasks
Focus Sessions
Distractions
Information Records
Reports
Progress

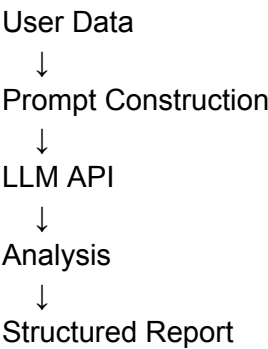
Analytics Engine

The analytics engine calculates:

- Focus Score
 - Focus session statistics
 - High-impact task statistics
 - Distraction statistics
 - Information consumption statistics
 - Weekly aggregates
-

AI Service

The AI service:



23. Complete Functional Requirements

Here is the consolidated list for your SRS.

ID	Functional Requirement
FR-01	User registration
FR-02	Registration validation
FR-03	Duplicate account prevention
FR-04	Secure credential storage
FR-05	User login
FR-06	Credential verification
FR-07	Session management
FR-08	Logout

FR-09 Display Day 1 questions

FR-10 Submit Day 1 answers

FR-11 Automatically save responses

FR-12 Edit responses

FR-13 Complete daily activity

FR-14 Create goals

FR-15 Categorize goals

FR-16 Edit/delete goals

FR-17 Prioritize goals

FR-18 Store goals

FR-19 Create actions

FR-20 Connect actions to goals

FR-21 Complete actions

FR-22 Display action roadmap

FR-23 Create tasks

FR-24 Record task time

FR-25 Identify task impact

FR-26 Analyze task impact

FR-27 Display high-impact tasks

FR-28 Start focus sessions

FR-29 Provide focus timer

FR-30 Record focus sessions

FR-31 Record interruptions

FR-32 Track distractions

- FR-33 Provide Say No scenarios
 - FR-34 Record user responses
 - FR-35 Provide optional AI feedback
 - FR-36 Record information activities
 - FR-37 Record information duration
 - FR-38 Record information usefulness
 - FR-39 Calculate information consumption
 - FR-40 Track 7-day progress
 - FR-41 Unlock next day after completion
 - FR-42 Calculate Focus Score
 - FR-43 Display score breakdown
 - FR-44 Collect 7-day data
 - FR-45 Send selected data to LLM API
 - FR-46 Generate AI analysis
 - FR-47 Generate weekly report
 - FR-48 Store AI report
 - FR-49 Display weekly report
 - FR-50 Generate Personal Focus Blueprint
 - FR-51 Store Focus Blueprint
 - FR-52 Allow users to view previous reports
 - FR-53 Start a new focus cycle
 - FR-54 Preserve previous cycle data
-

24. Non-Functional Requirements

These are also important for your university project.

Performance

NFR-01: Normal application pages should load within an acceptable response time under normal conditions.

NFR-02: Database queries should be optimized to avoid unnecessary delays.

Security

NFR-03: Passwords must not be stored as plain text.

NFR-04: API endpoints must verify user authentication where required.

NFR-05: Users should only be able to access their own personal data.

NFR-06: LLM API credentials must be stored securely on the server and never exposed in frontend JavaScript.

Usability

NFR-07: The interface should be simple enough for first-time users.

NFR-08: Users should clearly understand their current day and progress.

NFR-09: The application should be responsive on desktop and mobile screens.

Reliability

NFR-10: User responses should be saved reliably.

NFR-11: Temporary network/API failures should not cause previously saved answers to disappear.

Maintainability

NFR-12: Frontend, backend, database and AI services should be logically separated.

NFR-13: The application should use modular backend routes and controllers.

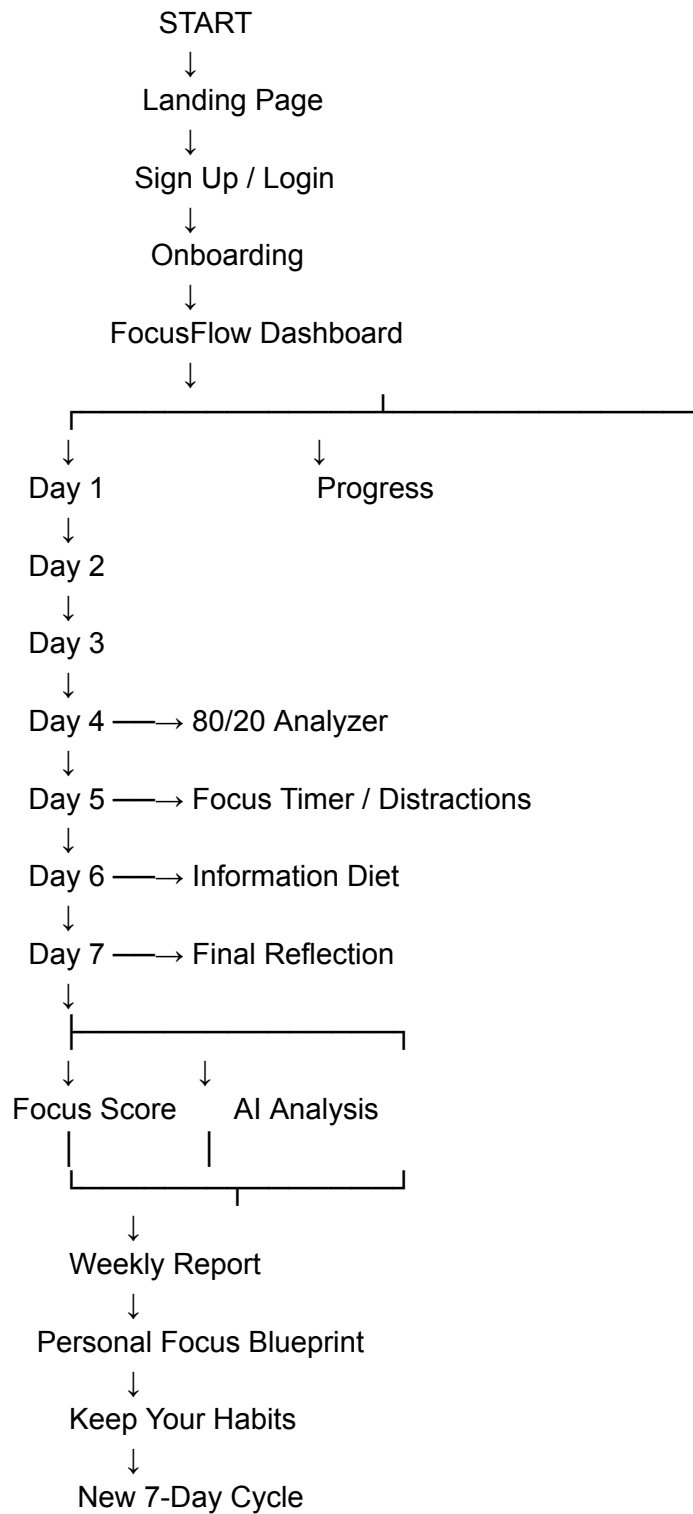
Privacy

NFR-14: Personal responses and reports should only be accessible to the corresponding user.

NFR-15: The system should clearly inform users when their data is sent to an external LLM service for analysis.

25. User Journey

The entire application can now be explained in one diagram:



26. MVP — What You Should Actually Build

This is **very important** for a university project.

Don't try to build every advanced feature shown in the workflow diagram immediately.

Phase 1 — Essential

Build these first:

- Registration/Login
- Dashboard
- Seven fixed days
- Day 1 questionnaire
- Goal management
- Goal → Action planner
- 80/20 Task Analyzer
- Distraction tracking
- Focus Timer
- Information Diet
- Daily progress
- MongoDB storage
- Focus Score
- AI report
- Weekly report
- Personal Focus Blueprint

Phase 2 — Additional

After the core system works:

- Say No Simulator
- More detailed analytics
- Focus history
- Previous cycle comparison
- Charts
- Better dashboard
- AI feedback on individual answers

Phase 3 — Future Scope

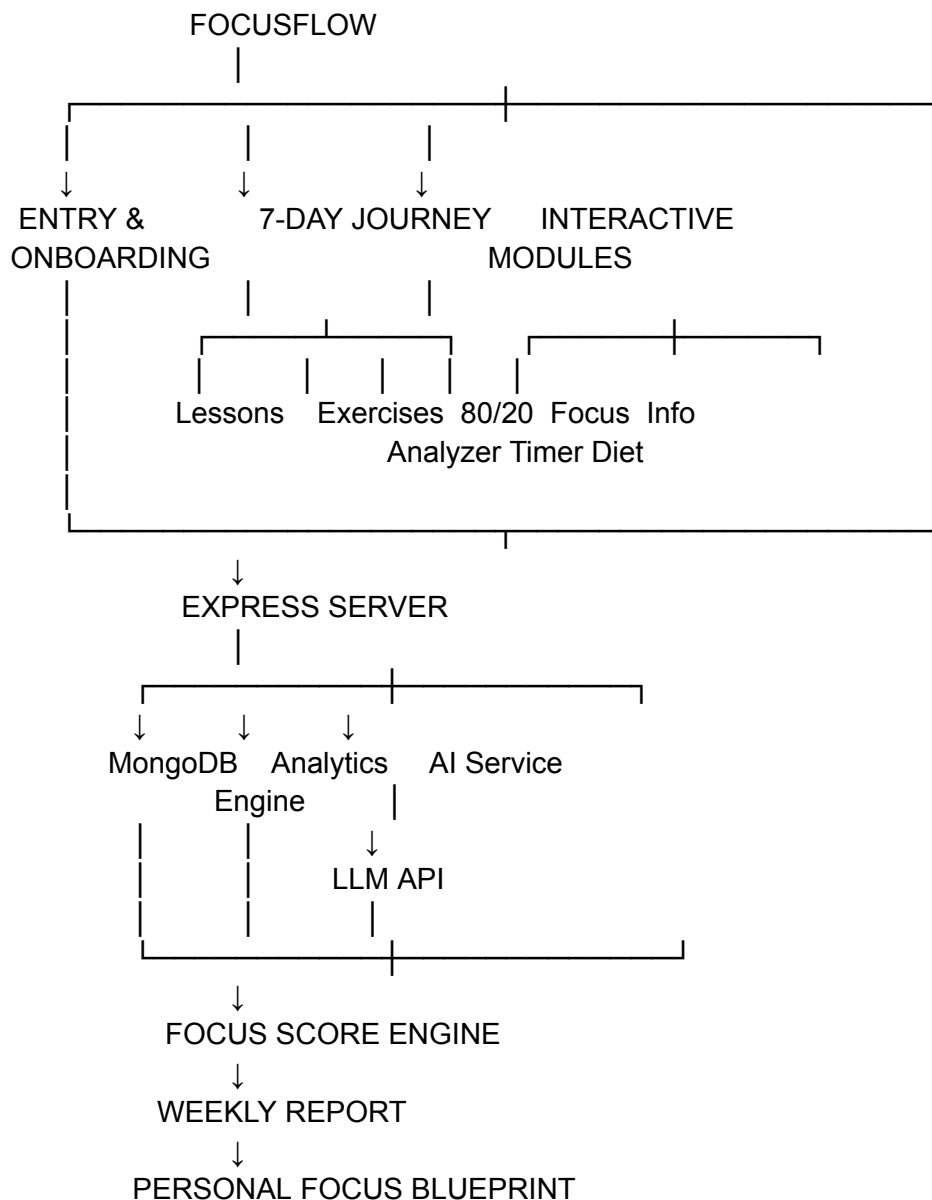
These can be mentioned in your report but don't need to be implemented:

- Browser extension for distraction tracking
- Automatic screen-time tracking

- Mobile application
- Calendar integration
- Smart notifications
- AI conversational focus coach
- Wearable/device integration
- Advanced behavioral analytics

27. Recommended Final System Architecture

Your original workflow can now be represented technically as:



28. The Core Value of FocusFlow

If your instructor asks:

"What makes this project different from simply reading the book?"

Your answer can be:

"The book provides the methodology, but FocusFlow provides an interactive system for applying, tracking, measuring, and reviewing that methodology. Instead of completing the exercises once on paper, users build a structured record of their goals, tasks, distractions, focus sessions, and reflections. The system calculates a transparent Focus Score and uses an LLM to analyze the collected information and generate a personalized weekly report and Focus Blueprint."

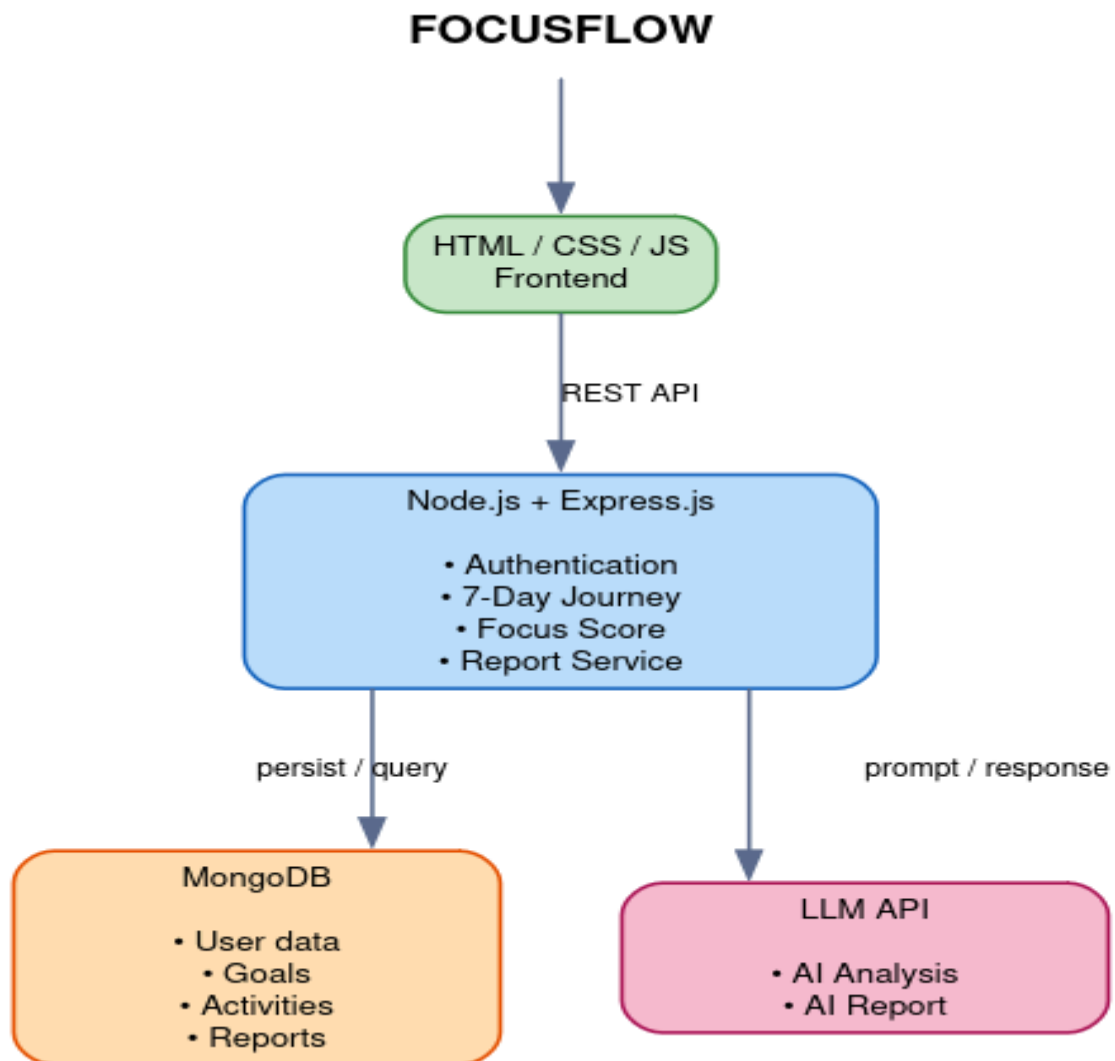
That is probably the **most important explanation** for your project.

And your complete concept can be summarized as:

Read → Do → Track → Measure → Analyze → Improve → Repeat

That should be the central philosophy of FocusFlow.

TECH STACK



```
focusflow/
├── backend/
│   ├── config/
│   │   └── db.js
│   ├── models/
│   │   ├── User.js
│   │   ├── Progress.js
│   │   ├── Goal.js
│   │   ├── Action.js
│   │   ├── Task.js
│   │   ├── FocusSession.js
│   │   ├── Information.js
│   │   ├── SayNo.js
│   │   └── Report.js
│   └── routes/
```

```
|
|
| | — auth.js
| | — days.js
| | — goals.js
| | — tasks.js
| | — focus.js
| | — info.js
| | — sayno.js
| | — report.js
| — middleware/
|   | — auth.js
| — services/
|   | — focusScore.js
|   | — gemini.js
| — utils/
|   | — promptBuilder.js
| — .env
| — server.js
| — package.json
| — frontend/
|   | — css/
|   |   | — style.css
|   | — js/
|   |   | — auth.js
|   |   | — dashboard.js
|   |   | — day1.js ... day7.js
|   |   | — focusTimer.js
|   |   | — charts.js
|   |   | — api.js
|   | — index.html      (Landing)
|   | — login.html
|   | — register.html
|   | — dashboard.html
|   | — day1.html ... day7.html
|   | — focus-session.html
|   | — report.html
|   | — blueprint.html
| — README.md
```